ECE4110 LAB TEST - COMPLETE CONFIGURATION GUIDE (BOLD COMMANDS

Example: Secure Access Configuration (Router R1)

Why: Configure console, VTY, and enable passwords to prevent unauthorized access. Apply sam

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```
R1> enable
R1# configure terminal
R1(config)# hostname R1
R1(config)# no ip domain-lookup
R1(config)# enable secret cisco123
R1(config)# line console 0
R1(config-line)# password class123
R1(config-line)# login
R1(config)# line vty 0 4
R1(config-line)# password remote123
R1(config-line)# login
R1(config)# banner motd #Unauthorized access is prohibited#
R1(config)# end
R1# copy running-config startup-config
```

Explanation:

- enable secret = secure EXEC password.
- line console 0 = protects console.
- line vty 0 4 = secures remote sessions.
- banner motd = unauthorized access warning.

Mode Prompt Reference

```
> = user EXEC mode
# = privileged EXEC mode
(config)# = global configuration mode
(config-if)# = interface configuration mode
(config-line)# = line configuration mode
Use 'end' or 'Ctrl+Z' to exit anytime.
```

Step 1 — Baseline Device Preparation

Why: Start clean, disable DNS lookup, and save configuration.

```
R1(config)# hostname R1
R1(config)# no ip domain-lookup
R1(config)# end
R1# copy running-config startup-config
```

Step 2 — VLAN & Inter-VLAN Routing (R1 & S1)

Why: Create VLANs for Students and Faculty, enable trunk link to R1.

```
S1(config)# vlan 10
S1(config-vlan)# name STUDENTS
S1(config)# vlan 20
S1(config-vlan)# name FACULTY
S1(config)# interface range fa0/2 - 10
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 10
S1(config)# interface range fa0/11 - 19
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 20
S1(config)# interface fa0/1
```

```
S1(config-if)# switchport mode trunk
S1(config-if)# switchport trunk allowed vlan 10,20
S1(config-if)# no shutdown
--------------------------------------------------------------
R1(config)# interface g0/0
R1(config-if)# no shutdown
R1(config)# interface g0/0.10
R1(config-subif)# encapsulation dot1Q 10
R1(config-subif)# ip address 192.168.10.1 255.255.255.0
R1(config)# interface g0/0.20
R1(config-subif)# encapsulation dot1Q 20
R1(config-subif)# ip address 192.168.20.1 255.255.255.0
--------------------------------------------------------------
Step 3 — DHCP Configuration (R2)
--------------------------------------------------------------
Why: Automate IP allocation for Department B; DNS linked to Admin DNS.
--------------------------------------------------------------
R2(config)# interface g0/0
R2(config-if)# ip address 192.168.30.1 255.255.255.0
R2(config-if)# no shutdown
R2(config)# ip dhcp excluded-address 192.168.30.1 192.168.30.10
R2(config)# ip dhcp pool DEPTB
R2(dhcp-config)# network 192.168.30.0 255.255.255.0
R2(dhcp-config)# default-router 192.168.30.1
R2(dhcp-config)# dns-server 192.168.100.11
--------------------------------------------------------------
Step 4 — Admin LAN Configuration (R3)
--------------------------------------------------------------
Why: Provide connectivity for DNS and Web servers.
--------------------------------------------------------------
R3(config)# interface g0/0
R3(config-if)# ip address 192.168.100.1 255.255.255.0
R3(config-if)# no shutdown
--------------------------------------------------------------
Step 5 — Serial WAN Links (R1–R2–R3)
--------------------------------------------------------------
Why: Enable backbone serial connections for routing.
--------------------------------------------------------------
R1(config)# interface s0/0/0
R1(config-if)# ip address 10.0.12.1 255.255.255.252
R1(config-if)# clock rate 64000
R1(config-if)# no shutdown
--------------------------------------------------------------
R2(config)# interface s0/0/0
R2(config-if)# ip address 10.0.12.2 255.255.255.252
R2(config-if)# no shutdown
R2(config)# interface s0/0/1
R2(config-if)# ip address 10.0.23.1 255.255.255.252
R2(config-if)# clock rate 64000
R2(config-if)# no shutdown
--------------------------------------------------------------
R3(config)# interface s0/0/0
R3(config-if)# ip address 10.0.23.2 255.255.255.252
R3(config-if)# no shutdown
--------------------------------------------------------------
Step 6 — OSPF Dynamic Routing
--------------------------------------------------------------
Why: Share routes dynamically between routers for all networks.
--------------------------------------------------------------
R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
```

```
R1(config-router)# network 192.168.10.0 0.0.0.255 area 0
R1(config-router)# network 192.168.20.0 0.0.0.255 area 0
R1(config-router)# network 10.0.12.0 0.0.0.3 area 0
--------------------------------------------------------------
R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 192.168.30.0 0.0.0.255 area 0
R2(config-router)# network 10.0.12.0 0.0.0.3 area 0
R2(config-router)# network 10.0.23.0 0.0.0.3 area 0
--------------------------------------------------------------
R3(config)# router ospf 1
R3(config-router)# router-id 3.3.3.3
R3(config-router)# network 192.168.100.0 0.0.0.255 area 0
R3(config-router)# network 10.0.23.0 0.0.0.3 area 0
--------------------------------------------------------------
Step 7 — ACL Configuration (R3)
--------------------------------------------------------------
Why: Restrict student access to Admin Web; allow Faculty traffic.
--------------------------------------------------------------
R3(config)# ip access-list extended BLOCK_STUDENTS_ALL
R3(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 host 192.168.100.10 eq 80
R3(config-ext-nacl)# deny tcp 192.168.10.0 0.0.0.255 host 192.168.100.10 eq 443
R3(config-ext-nacl)# deny tcp host 192.168.30.12 host 192.168.100.10 eq 80
R3(config-ext-nacl)# deny tcp host 192.168.30.12 host 192.168.100.10 eq 443
R3(config-ext-nacl)# deny tcp host 192.168.30.14 host 192.168.100.10 eq 80
R3(config-ext-nacl)# deny tcp host 192.168.30.14 host 192.168.100.10 eq 443
R3(config-ext-nacl)# permit ip any any
R3(config)# interface s0/0/0
R3(config-if)# ip access-group BLOCK_STUDENTS_ALL in
--------------------------------------------------------------
Step 8 — DNS and Web Server Configuration (Admin Network)
--------------------------------------------------------------
Why: Enable www.mehrab.ac.bd and DNS name resolution in Admin LAN.
--------------------------------------------------------------
DNS Server (192.168.100.11)
Desktop → IP Config:
  IP: 192.168.100.11
  Mask: 255.255.255.0
  Gateway: 192.168.100.1
  DNS: 192.168.100.11
Services → DNS:
  Turn ON
  Add Record: Name: www.mehrab.ac.bd → Address: 192.168.100.10
--------------------------------------------------------------
Web Server (192.168.100.10)
Desktop → IP Config:
  IP: 192.168.100.10
  Mask: 255.255.255.0
  Gateway: 192.168.100.1
  DNS: 192.168.100.11
Services → HTTP: Turn ON (optional HTTPS)
Edit Index.html:
  <h1>Welcome to Admin Web Portal</h1>
  <p>This site belongs to www.mehrab.ac.bd</p>
--------------------------------------------------------------
Verification:
PC> ping www.mehrab.ac.bd
PC> Web Browser → www.mehrab.ac.bd
  Faculty = Access Allowed
  Student = Request Timeout (Blocked)
--------------------------------------------------------------
```

Verification Commands and IP Table

```
S1# show vlan brief
S1# show interfaces trunk
R2# show ip dhcp binding
R1# show ip ospf neighbor
R2# show ip ospf neighbor
R3# show ip ospf neighbor
R3# show access-lists BLOCK_STUDENTS_ALL
R3# show ip interface serial0/0/0 | include access
```

Dept	Network/Subnet	Gateway IP	Notes
Dept-A VLAN10	192.168.10.0/24	192.168.10.1	Students VLAN
Dept-A VLAN20	192.168.20.0/24	192.168.20.1	Faculty VLAN
Dept-B LAN	192.168.30.0/24	192.168.30.1	DHCP on R2
Admin LAN	192.168.100.0/24	192.168.100.1	DNS & Web Servers
WAN R1-R2	10.0.12.0/30	R1=10.0.12.1, R2=10.0.12.2	Serial Link
WAN R2-R3	10.0.23.0/30	R2=10.0.23.1, R3=10.0.23.2	Serial Link

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